

Spiking Reservoir State-Space Encoding for Perturbation-Characterized Affective EEG Signal Analysis

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Abstract—Affective electroencephalographic (EEG) event-related potentials (ERPs) are noisy, subject-dependent, and only partially observed projections of neural dynamics. Endpoint accuracy alone does not reveal what temporal information a representation retains or discards, and the perturbation behavior of spiking-reservoir encodings for affective ERP has not been systematically reported. We define a fixed leaky integrate-and-fire (LIF) reservoir with a six-bin spike-count (BSC₆) code as a temporal encoding operator, compress it for a linear readout, and audit it under subject-grouped validation against a classical EEG/ERP baseline and under amplitude-noise, temporal-jitter, and channel-dropout perturbations. The reservoir embedding does not exceed the classical baseline on clean classification (balanced accuracy 0.463 vs. 0.485) yet degrades more gradually under representation-level amplitude noise and channel dropout. A fold-local principal-component analysis shows no evidence of optimistic inflation from the transductive basis (balanced accuracy 0.465 vs. 0.463).

Index Terms—EEG, event-related potentials, spiking neural networks, reservoir computing, biomedical signal processing, affective computing, perturbation analysis.

I. INTRODUCTION

Electroencephalographic event-related potentials (ERPs) are time-locked measurements of neural response dynamics that offer millisecond-scale temporal information for biomedical data analysis, yet the observed waveform is a noisy, low-dimensional, and subject-dependent projection of an unobserved neural dynamical system [12], [13]. Inter-subject variability, trial averaging, volume conduction, and measurement noise can cause endpoint classifiers to conflate genuine temporal structure with subject-specific nuisance variation, so a representation that performs well on one clean split may not preserve the same signal properties when amplitude noise, temporal misalignment, or channel loss is introduced.

Reporting only endpoint accuracy does not reveal *which* temporal features a representation preserves or suppresses. This gap is acute for spiking encodings: fixed recurrent reservoirs and spiking networks are expressive nonlinear temporal expansions [1], [4], but in EEG/ERP work they are most often evaluated as end-to-end classifiers [9], [15] rather than characterized as temporal operators whose discrete spike-count observables remain traceable to post-stimulus time windows.

The perturbation behavior of affective-ERP reservoir encodings—their response to amplitude noise, temporal jitter, and channel dropout—is rarely reported, even though

preprocessing and acquisition choices materially affect EEG representations [16]–[18].

This paper studies a fixed spiking reservoir state-space encoding for affective ERP signal analysis and audits it as a temporal encoding operator (Fig. 1). Each ERP observation is mapped through a fixed leaky integrate-and-fire (LIF) reservoir, summarized by six post-onset spike-count bins (BSC₆), and compressed for a linear readout. The contributions are:

- 1) A fixed LIF/BSC₆ temporal encoding operator for affective EEG/ERP observations, defined as a composition of a nonlinear reservoir map, spike-count binning, and an unsupervised projection (Sec. III, Eq. (5)).
- 2) Evaluation of the reservoir embedding against a classical EEG/ERP baseline under subject-grouped validation, using balanced accuracy, macro-F1, and macro one-vs-rest AUC (Sec. V-A, Table II).
- 3) Perturbation-response characterization under amplitude noise, temporal jitter, and channel dropout, separating representation-level corruption from raw-signal recomputation (Sec. V-B, Table IV, Fig. 3).
- 4) A fold-local PCA sensitivity analysis bounding the transduction of the global unsupervised projection (Sec. V-A, Table III).

Clinical labels and affective categories define the external task structure; the reservoir-based temporal representation is the object of study.

II. RELATED WORK

ERP and EEG analysis have traditionally relied on time-window amplitudes and latencies, time-domain descriptors such as Hjorth parameters [10], spectral and wavelet summaries [11], and linear statistical models [13]. These features are interpretable and correspond to known components such as the P300 and late positive potential, and source-level analyses show that evoked responses carry structured temporal dynamics [12]. Fixed handcrafted windows may, however, underrepresent nonlinear temporal interactions and can be sensitive to subject-specific variability.

Reservoir computing offers an alternative in which a high-dimensional recurrent dynamical system transforms an input sequence into a separable state trajectory [1]–[4]. In liquid state machines, recurrent spiking dynamics induce a nonlinear temporal basis on which a simple readout is trained; related

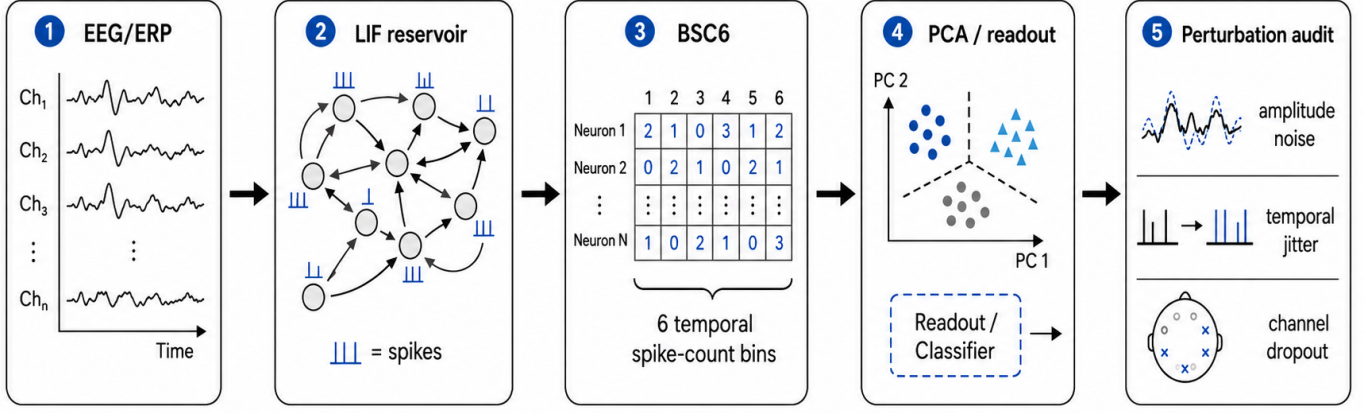


Fig. 1. Reservoir encoding and perturbation-audit pipeline. Trial-averaged affective EEG/ERP observations are mapped through a fixed LIF reservoir, summarized by six temporal spike-count bins (BSC_6), compressed for linear readout, and evaluated under amplitude noise, temporal jitter, and channel-dropout perturbations.

spiking models encode information in spike timing and counts [5], [6]. This separation of fixed dynamics from a trained readout is attractive for biomedical signal analysis because the reservoir can be studied as a measurement operator rather than as a fully optimized black-box classifier.

Spiking and deep networks have been applied to EEG and ERP signals [9], [14], [15], but most studies emphasize end-point accuracy, while preprocessing and acquisition choices that strongly affect EEG representations are often under-examined [18]. For perturbation-sensitive physiological data, accuracy alone is insufficient: a representation should also be assessed through subject-level generalization, controlled corruption, and representation-geometry analysis [19]. Prior work in the ARSPI-Net line developed neuromorphic reservoir feature extraction for affective EEG [7], [8]; the present paper isolates and audits the spiking temporal-encoding component as a measurement operator.

III. METHODS

A. ERP Observation Model

Let $X_s^{(c)} \in \mathbb{R}^{C \times T}$ denote the trial-averaged ERP observation for subject s and affective condition c , with C electrodes (channels) and T temporal samples; observations are stored as $T \times C$ arrays and transposed before encoding. We treat $X_s^{(c)}$ as a noisy observation of an unobserved neural state trajectory:

$$\xi_{t+1} = f_\theta(\xi_t, u_t) + \eta_t, \quad x_t = H\xi_t + \epsilon_t, \quad (1)$$

where ξ_t is the latent neural state, H is a measurement operator induced by scalp observation, and ϵ_t captures measurement and preprocessing noise.

B. Spiking Reservoir Encoding

For each electrode trace, the signal is passed through a fixed LIF reservoir of 256 neurons with recurrent weights W (spectral radius $\rho = 0.9$), input weights W_{in} , leak parameter

$\beta = 0.05$, and firing threshold ϑ . A compact discrete-time representation is

$$v_{t+1} = \beta v_t + W z_t + W_{\text{in}} x_t - \vartheta r_t, \quad (2)$$

$$z_{t+1} = \mathbf{1}[v_{t+1} \geq \vartheta], \quad (3)$$

where v_t is the membrane state, z_t is the binary spike vector, and r_t denotes reset after threshold crossing. The reservoir is fixed during feature extraction; only the downstream readout is trained.

Post-onset activity is summarized by six equal temporal bins. For reservoir unit j and bin k , the binned spike-count code is

$$b_{j,k} = \sum_{t \in \mathcal{B}_k} z_j(t), \quad k = 1, \dots, 6. \quad (4)$$

using a post-onset window $t \in [10, 70]$. This produces a BSC_6 vector that preserves approximate temporal traceability while reducing sensitivity to sample-level fluctuations. Per-electrode spike-bin features are compressed to 64 principal components, and the electrode-level embeddings are concatenated across the 34 channels into a trial-level reservoir representation $E_s^{(c)} \in \mathbb{R}^{2176}$. This principal-component projection is estimated once, unsupervised and without affective labels, over the spike-bin vectors pooled across all observations and electrodes of the full cohort. Because the basis is fitted on every subject—including those later held out during validation—it is transductive with respect to subject inclusion, and we therefore report it as a fixed representation audit rather than as leakage-free preprocessing. To bound any optimism introduced by this fixed basis, we additionally evaluate a fold-local variant in which the projection is refitted on each fold’s training subjects only (Sec. V-A).

Writing $X \in \mathbb{R}^{C \times T}$ for a trial-averaged observation with C channels and T samples, and $s_t = z_t \in \{0, 1\}^N$ for the reservoir spike state at post-onset sample t , the encoding is a

fixed composition of a nonlinear temporal map, spike-count binning, and the unsupervised projection:

$$b_k = \sum_{t \in \mathcal{T}_k} s_t, \quad k = 1, \dots, 6, \quad (5)$$

$$E = \phi_{\text{PCA}}([b_1, \dots, b_6]) = \Phi(X),$$

so that $E = \Phi(X)$ is a fixed, label-free observable of the input.

Fig. 2 shows the reservoir-derived computational objects for the three affective conditions: the LIF spike raster, the membrane-potential response, and the BSC_6 embedding projected to two principal components. Spike density and membrane dynamics are organized in post-onset time; the projection summarizes condition separation at the population level.

C. Baselines and Readout

The primary baseline consists of bandpower and ERP-derived features evaluated with the same subject-grouped validation protocol. A linear readout is used to test representational separability without confounding the result with a high-capacity classifier. Decision-tree and random-forest readouts are reserved for sensitivity checks.

D. Perturbation Diagnostics

Two perturbation regimes are separated. Representation-level perturbations corrupt the extracted feature vector and test whether downstream classifiers are sensitive to feature-space noise and dropout. Raw-signal recomputation applies perturbations before feature extraction and recomputes the reservoir representation. The second regime is stricter because corruption propagates through reservoir dynamics before classification. Formally, a perturbation $\mathcal{P}_\eta$ is applied either to the signal, $X \mapsto \mathcal{P}_\eta(X)$ with $E = \Phi(\mathcal{P}_\eta(X))$, or to the representation, $E \mapsto \mathcal{P}_\eta(E)$, where η indexes the amplitude-noise signal-to-noise ratio, the temporal-jitter offset, or the channel-dropout fraction.

The perturbations are:

- additive amplitude noise at multiple signal-to-noise ratios;
- temporal jitter of ERP alignment windows;
- channel dropout at increasing electrode-removal rates.

Performance is reported as balanced accuracy (BA) and macro-F1.

IV. EXPERIMENTAL PROTOCOL

A. Dataset and Task

The experiments use a trial-averaged affective ERP dataset with 211 subjects after exclusion of one anomalous subject record. Each subject contributes three affective conditions: negative, neutral, and pleasant, yielding 633 subject-condition observations. The ERP traces contain 34 scalp channels downsampled to 256 Hz, with 256 post-stimulus samples per epoch, and are baseline-corrected relative to stimulus onset. Subject-level data are maintained in a restricted research environment; only aggregate, de-identified results are reported here. Table I summarizes the dataset and evaluation protocol.

TABLE I
DATASET AND EVALUATION PROTOCOL SUMMARY.

Item	Value
Subjects (after exclusion)	211
Excluded records	1 (recording anomaly)
Observations	633 (subject $\times$ condition)
Conditions	3 (negative, neutral, pleasant)
Channels	34
Sampling rate	256 Hz
Post-stimulus samples	256
Reservoir embedding dim. (E)	2176
Validation	subject-grouped cross-validation

TABLE II
CLEAN SUBJECT-LEVEL AFFECTIVE CONDITION PERFORMANCE.

Configuration	Feature family	BA	Macro-F1
A0	Classical bandpower/ERP	0.485	0.483
A1	Reservoir embedding E	0.463	0.460

The three-class affective condition task tests whether the temporal representation separates ERP responses under subject-level validation.

B. Validation

All classifier evaluations use subject-level partitioning: observations from the same subject do not appear simultaneously in training and test folds. Reported metrics include balanced accuracy and macro-F1. For provenance, the clean A0 and A1 results, the A1 representation-level perturbations, and the fold-local PCA sensitivity were recomputed under this protocol; the raw-signal recomputation values and the baseline (A0) perturbation values are carried from prior validated aggregate outputs of the same pipeline and are not re-derived here.

C. Manuscript Boundary

The present paper isolates the reservoir temporal-encoding component. Graph-topological observables, structure–function coupling, and closed-loop evidence-accumulation analyses belong to other parts of the doctoral framework and are outside this submission.

V. RESULTS

A. Clean Subject-Level Classification

Table II reports clean three-class affective condition performance for the classical baseline (A0) and the reservoir embedding (A1), both reproduced under the present subject-grouped protocol. Under clean static classification, conventional ERP-derived features remain competitive with the reservoir embedding.

The reservoir does not outperform the classical baseline under clean conditions (Table II), so clean accuracy alone does not justify the spiking encoding.

Fold-local PCA sensitivity. Because the unsupervised projection of Eq. (5) is fitted once over the pooled cohort, we test whether its transduction inflates performance by refitting it on each fold’s training subjects only (Table III). The fold-local

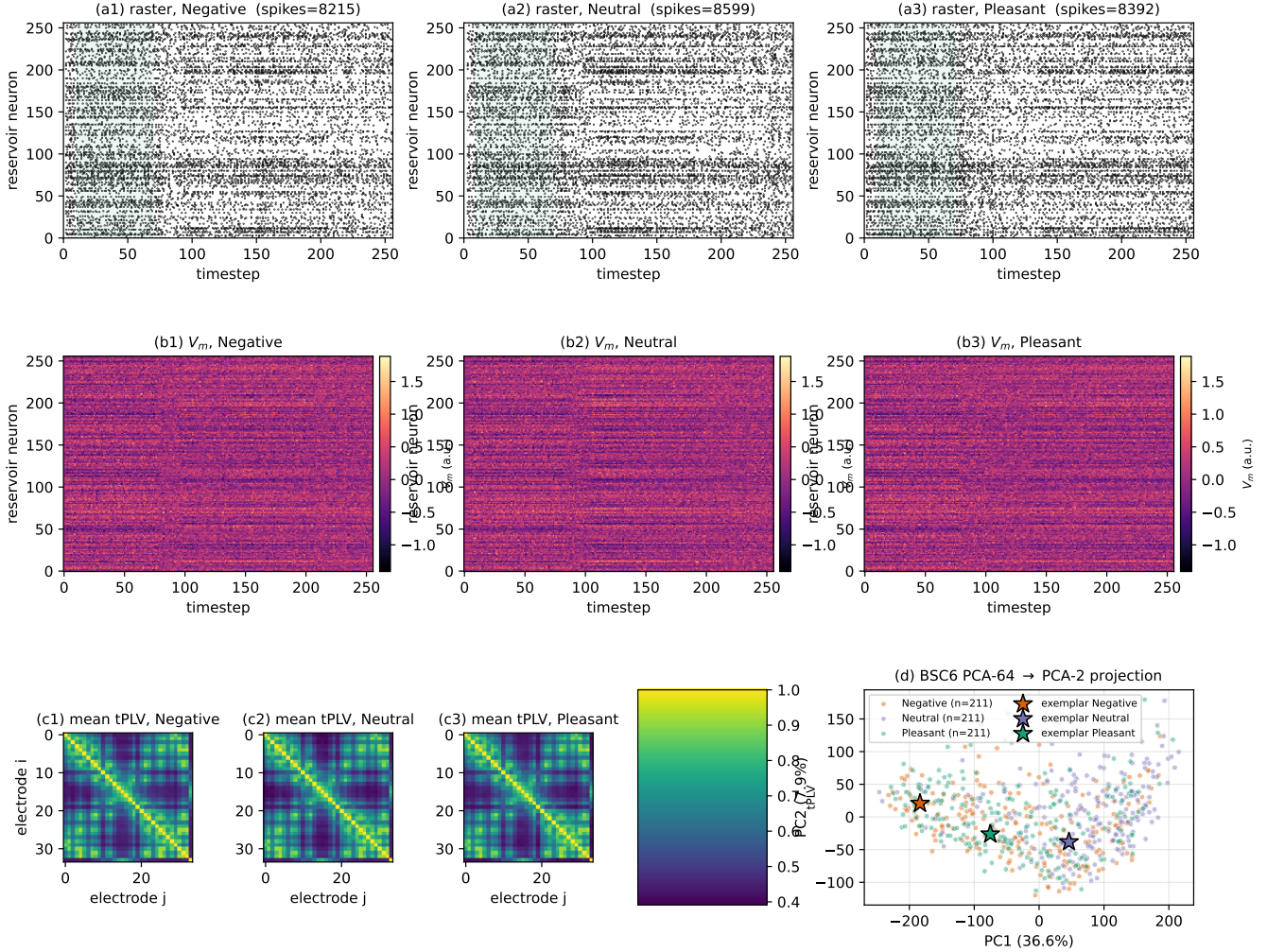


Fig. 2. Reservoir-derived computational objects for affective ERP observations. (a) LIF spike raster. (b) Membrane response. (d) BSC₆ embedding projected to the first two principal components.

TABLE III
FOLD-LOCAL PCA SENSITIVITY FOR THE RESERVOIR EMBEDDING E (A1) UNDER THE SUBJECT-GROUPED PROTOCOL. ROC-AUC IS MACRO ONE-VS-REST FROM POOLED OUT-OF-FOLD PROBABILITIES.

PCA basis	BA	Macro-F1	OvR-AUC
Global pooled (transductive)	0.463	0.460	0.644
Fold-local (per-fold train)	0.465	0.462	0.649

variant yields balanced accuracy 0.465, macro-F1 0.462, and macro one-vs-rest AUC 0.649, compared with 0.463, 0.460, and 0.644 for the fixed global basis. The two projections produce indistinguishable performance under the tested protocol.

B. Representation-Level Perturbation Response

Table IV summarizes feature-space perturbation results. Under additive amplitude corruption, the classical bandpower baseline degrades from BA 0.485 to 0.395 at 5 dB. The reservoir embedding degrades from BA 0.463 to 0.449. A similar pattern holds under feature dropout: the reservoir embedding remains close to its clean value at 30% dropout while the classical baseline drops to 0.408.

The spike-bin representation shows a different sensitivity profile from the classical baseline when corruption is applied after feature extraction: it is less sensitive to additive feature noise and dropout in this regime.

C. Raw-Signal Recomputed Perturbations

A stricter test corrupts the signal before feature extraction and recomputes the representation. Table V reports recomputa-

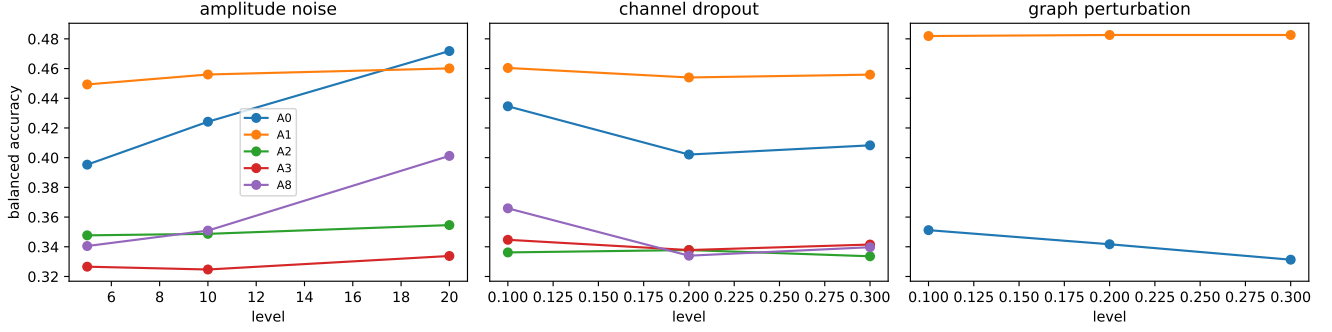


Fig. 3. Perturbation response under controlled corruption. Balanced accuracy as a function of perturbation level for the reservoir embedding E (A1) and the classical bandpower/ERP baseline (A0) under amplitude noise and channel dropout.

TABLE IV
REPRESENTATION-LEVEL PERTURBATION DIAGNOSTICS.

Perturbation	Level	A0 BA	A1 BA	Difference
Clean	—	0.485	0.463	-0.022
Amplitude noise	5 dB	0.395	0.449	+0.054
Channel dropout	30%	0.408	0.456	+0.048

TABLE V
RAW-SIGNAL RECOMPUTATION PERTURBATION DIAGNOSTIC.

Condition	BA	Macro-F1
Clean recomputation subset	0.433	0.426
Temporal jitter, ± 50 ms	0.389	0.352
Amplitude noise, 5 dB	0.429	0.419
Channel dropout, 30%	0.362	0.233

tion diagnostics. Temporal jitter and channel dropout produce larger degradation than the representation-level regime. This is expected: the reservoir state trajectory depends on the full temporal input sequence, so misalignment and channel removal alter the dynamical drive before spike counts are formed.

The two regimes produce different outcomes. The reservoir embedding is stable when the representation is corrupted after extraction, but its performance degrades when perturbations alter the temporal drive that determines reservoir state evolution. Preprocessing alignment and electrode availability remain critical.

The degradation curves in Fig. 3 show the same separation: under amplitude noise and channel dropout, the reservoir embedding (A1) maintains a flatter balanced-accuracy profile than the classical baseline (A0).

VI. DISCUSSION

Conventional ERP-derived features remain competitive under clean static classification (Table II). The perturbation results, however, separate two operating regimes. When corruption is applied after feature extraction, the spike-bin embedding retains balanced accuracy more gradually than the

classical baseline (Table IV, Fig. 3). When corruption enters before extraction—temporal jitter and channel removal that alter the reservoir’s input trajectory—performance degrades more sharply (Table V). The encoding is therefore robust to post-hoc feature corruption but dependent on temporal alignment and electrode availability at acquisition time.

The fold-local PCA sensitivity analysis (Table III) confirms that the cohort-pooled projection does not inflate the reported reservoir-embedding performance under the tested protocol. The affective labels serve as an external task structure for testing separability; no diagnostic claim is made.

Several scope boundaries remain. The dataset is a single affective ERP cohort, and the results require external validation before generalization to other EEG paradigms. The study evaluates software-level representations, not neuromorphic hardware energy or latency, which would require measurement on dedicated platforms [20], [21]. The perturbation diagnostics are controlled tests, not substitutes for full acquisition-artifact conditions. Graph topology and structure–function coupling are not evaluated here.

VII. CONCLUSION

Under clean static classification, conventional ERP features remain competitive. The reservoir embedding defines a non-linear temporal encoding with a distinct perturbation-response profile: it degrades more gradually than the classical baseline under representation-level amplitude noise and channel dropout, while remaining sensitive to temporal misalignment and channel loss when perturbations enter before feature extraction. Each BSC_6 component corresponds to a post-onset spike-count interval, and the fold-local PCA analysis confirms that the transductive basis does not inflate reported performance. The operating regime of the spiking encoding depends on the perturbation type and entry point.

A vector camera-ready version of the pipeline figure should replace the current raster asset. Fold-local and full within-fold recomputation should extend to all descriptor streams beyond the embedding evaluated here. Multi-cohort and multi-paradigm validation is required before the operating-regime

characterization generalizes. Any claim of neuromorphic efficiency requires direct hardware latency and energy measurement, which this software study does not provide.

Data Availability and Scope. The EEG/ERP data used in this study are access-controlled research data and are not redistributed with this submission. Analyses are reported only as aggregate, de-identified results. Clinical labels, where present in the source cohort, are used only to define biomedical signal-analysis context and are not interpreted as diagnostic decisions.

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